Finite Element Method

Strong and Weak Forms: 1D Problems

MECH4103 — Spring 2026

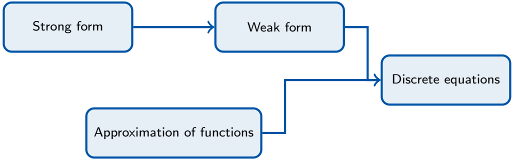
Dr. Behrouz Arash
Associate Professor in Mechanical Engineering

Oslo Metropolitan University

January 3, 2026

Roadmap

- A roadmap for the development of the finite element method is shown in the figure.
- There are three distinct ingredients that are combined to arrive at the discrete equations, which are then solved by a computer:
  1. The **strong form**: governing equations and boundary conditions.
  2. The **weak form**.
  3. The **approximation functions**.
- The approximation functions are combined with the weak form to obtain the **discrete finite element equations**.



Weak Forms

- In the preceding lecture, we stated the governing equations for elasticity and field problems in terms of partial differential equations.
- We consider an approach called the **weak form** of the differential equations.
- In the next lectures, we also consider a related but alternate approach based on a **variational approach**.
- Later we show that the two approaches are closely related and often may be used interchangeably.
- A weak form is obtained using the following steps:
  - Multiply each equation by an appropriate *arbitrary* function.
  - Integrate this product over the space domain of the problem.
  - Use **integration by parts** to reduce the order of derivatives to a minimum.
  - Introduce boundary conditions if possible.

- The **strong form** for 1D stress analysis:

$$
\begin{aligned}
&\frac{\mathrm{d}}{\mathrm{d}x}\left(AE\frac{\mathrm{d}u}{\mathrm{d}x}\right) + b = 0, \quad 0 < x < l, \\
&\sigma n = En\frac{\mathrm{d}u}{\mathrm{d}x} = \bar{t} \quad \text{on } \Gamma_t, \\
&u = \bar{u} \quad \text{on } \Gamma_u.
\end{aligned} \tag{1}
$$

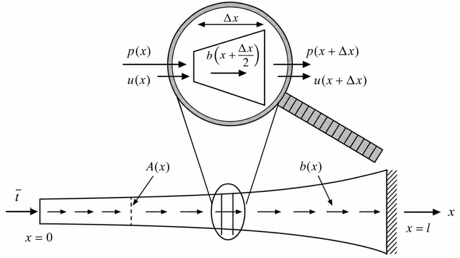


Figure 3.2 A one-dimensional stress analysis (elasticity) problem.

- Equilibrium of a differential element gives:

$$
-p(x) + b\left(x + \frac{\Delta x}{2}\right)\Delta x + p(x + \Delta x) = 0.
$$

- As $\Delta x \to 0$:

$$
\frac{\mathrm{d}p(x)}{\mathrm{d}x} + b(x) = 0.
$$

- Using Hooke's law, the internal force is:

$$
\sigma(x) = E(x)\,\varepsilon(x), \quad \varepsilon(x) = \frac{\mathrm{d}u}{\mathrm{d}x},
$$

$$
p(x) = A(x)\,\sigma(x).
$$

- Substituting into the equilibrium equation:

$$
\frac{\mathrm{d}}{\mathrm{d}x}\left(AE\frac{\mathrm{d}u}{\mathrm{d}x}\right) + b = 0, \quad 0 < x < l.
$$

- We first recall the formula for **integration by parts**:

$$\int_\Omega w \frac{\mathrm{d}f}{\mathrm{d}x}\,\mathrm{d}x = (wfn)\big|_\Gamma - \int_\Omega f \frac{\mathrm{d}w}{\mathrm{d}x}\,\mathrm{d}x = (wfn)\big|_{\Gamma_u} + (wfn)\big|_{\Gamma_t} - \int_\Omega f \frac{\mathrm{d}w}{\mathrm{d}x}\,\mathrm{d}x. \tag{2}$$

  Here Ω denotes the 1D domain (e.g. $[0, l]$); Γ denotes all boundary points; Γ_u and Γ_t are the prescribed-displacement and traction boundaries, respectively. Weight functions satisfy $w = 0$ on Γ_u; trial solutions satisfy $u = \bar{u}$ on Γ_u.

- Multiply the first two equations of the strong form (1) by w and integrate:

$$\begin{aligned} &\text{(a)} \quad \int_\Omega w\left(\frac{\mathrm{d}}{\mathrm{d}x}\left(AE\frac{\mathrm{d}u}{\mathrm{d}x}\right) + b\right)\mathrm{d}x = 0 \qquad \forall w, \\ &\text{(b)} \quad \big(wA(\bar{t} - \sigma n)\big)\big|_{\Gamma_t} = 0 \qquad \forall w. \end{aligned} \tag{3}$$

- Integrating (a) by parts and combining with (b):

$$(wA\sigma n)\big|_{\Gamma_u} + (wA\bar{t})\big|_{\Gamma_t} - \int_\Omega \frac{\mathrm{d}w}{\mathrm{d}x} AE \frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x + \int_\Omega wb\,\mathrm{d}x = 0 \qquad \forall w,\ w = 0 \text{ on } \Gamma_u. \tag{4}$$

- The boundary term on Γ_u vanishes because $w\big|_{\Gamma_u} = 0$. The **weak form** then becomes:

Weak Form — 1D Elasticity

Find $u(x) \in U$ such that

$$\int_{\Omega} \frac{\mathrm{d}w}{\mathrm{d}x} AE \frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x = (wA\bar{t})\big|_{\Gamma_t} + \int_{\Omega} wb\,\mathrm{d}x \qquad \forall w \in U_0. \tag{5}$$

- The set of admissible **trial solutions**:

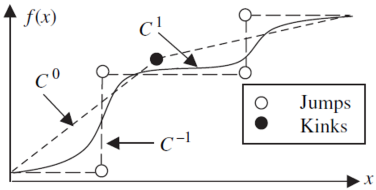
$$U = \{u(x)\,|\,u(x) \in H^1,\ u = \bar{u} \text{ on } \Gamma_u\}.$$

- H^1 is the **Sobolev space** of order 1: functions $u(x)$ with $\int u^2\,\mathrm{d}x < \infty$ and $\int (du/dx)^2\,\mathrm{d}x < \infty$. All H^1 functions are C^0 continuous.
- The set of admissible **weight functions**:

$$U_0 = \{w(x)\,|\,w(x) \in H^1,\ w = 0 \text{ on } \Gamma_u\}.$$

- U_0 is identical to U, except that weight functions must vanish on the essential boundary.

- C^{-1}: can have both **kinks** and **jumps**.
- C^{0}: no jumps (discontinuities), but has kinks.
- C^{1}: no kinks or jumps.
- There is a progression of smoothness as the superscript increases.



Examples of C^{-1}, C^0 and C^1 functions.

Smoothness	Kinks	Jumps	Comments
C^{-1}	Yes	Yes	Piecewise continuous
C^{0}	Yes	No	Piecewise cont. diff.
C^{1}	No	No	Continuously diff.

- **Strong form** for 1D heat conduction:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(Ak\frac{\mathrm{d}T}{\mathrm{d}x}\right)+s=0 \quad \text{on } \Omega,$$
$$qn=-kn\frac{\mathrm{d}T}{\mathrm{d}x}=\bar{q} \quad \text{on } \Gamma_q, \tag{6}$$
$$T=\bar{T} \quad \text{on } \Gamma_T.$$

- Γ_T: **essential (Dirichlet)** boundary — prescribed temperature.
- Γ_q: **natural (Neumann)** boundary — prescribed flux.

- Multiply the first two equations by w and integrate over Ω:

$$\text{(a)}\ \int_\Omega w\frac{\mathrm{d}}{\mathrm{d}x}\left(Ak\frac{\mathrm{d}T}{\mathrm{d}x}\right)\mathrm{d}x+\int_\Omega ws\,\mathrm{d}x=0 \quad \forall w,$$
$$\text{(b)}\ \left(wA(qn-\bar{q})\right)\big|_{\Gamma_q}=0 \quad \forall w. \tag{7}$$

- Integration by parts of (a):

$$\int_\Omega \frac{\mathrm{d}w}{\mathrm{d}x}Ak\frac{\mathrm{d}T}{\mathrm{d}x}\,\mathrm{d}x=\left(wAk\frac{\mathrm{d}T}{\mathrm{d}x}n\right)\Big|_\Gamma$$
$$+\int_\Omega ws\,\mathrm{d}x \quad \forall w,\ w{=}0 \text{ on } \Gamma_T. \tag{8}$$

- Recalling $w = 0$ on Γ_T and combining with (7b), the **weak form** is:

Weak Form — 1D Heat Conduction

Find $T(x) \in U$ such that

$$\int_\Omega \frac{\mathrm{d}w}{\mathrm{d}x} Ak \frac{\mathrm{d}T}{\mathrm{d}x}\,\mathrm{d}x = -(wA\bar{q})\big|_{\Gamma_q} + \int_\Omega ws\,\mathrm{d}x \qquad \forall w \in U_0. \tag{9}$$

- Notice the structural similarity between (9) and the elasticity weak form (5).
- The equations for heat conduction, diffusion, and elasticity are all of the **general form**:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(A\kappa\frac{\mathrm{d}\theta}{\mathrm{d}x}\right) + f = 0 \quad \text{on } \Omega. \tag{10}$$

- The table below gives the meaning of each variable for several applications:

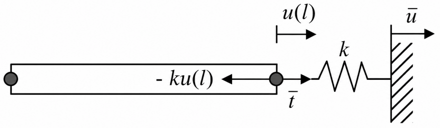
Field/parameter	Elasticity	Heat conduction	Diffusion
θ	u	T	c
κ	E	k	k
f	b	s	s
$\bar{\Phi}$	$\bar{t}$	$-\bar{q}$	$-\bar{q}$
$\bar{\theta}$	u	T	c
Γ_Φ	Γ_t	Γ_q	Γ_q
Γ_θ	Γ_u	Γ_T	Γ_c
β	k	h	h

- The natural boundary conditions can be **generalized** as:

$$\left(\kappa n \frac{\mathrm{d}\theta}{\mathrm{d}x} - \bar{\Phi}\right) + \beta(\theta - \bar{\theta}) = 0 \quad \text{on } \Gamma_\Phi. \tag{11}$$

- When $\beta = 0$: reduces to the standard natural BC.
- When $\beta \to \infty$: recovers the essential BC (penalty interpretation).
- Example: elastic bar with a spring attached at $x = l$ ($\beta = k$):

$$\left(E(l)n(l)\frac{\mathrm{d}u}{\mathrm{d}x}(l) - \bar{t}\right) + k\big(u(l) - \bar{u}\big) = 0.$$



- **General strong form** for 1D problems:

$$
\begin{aligned}
&\text{(a)} \quad \frac{\mathrm{d}}{\mathrm{d}x}\left(A\kappa\frac{\mathrm{d}\theta}{\mathrm{d}x}\right) + f = 0 \quad \text{on } \Omega, \\
&\text{(b)} \quad \left(\kappa n\frac{\mathrm{d}\theta}{\mathrm{d}x} - \bar{\Phi}\right) + \beta(\theta - \bar{\theta}) = 0 \quad \text{on } \Gamma_{\Phi}, \\
&\text{(c)} \quad \theta = \bar{\theta} \quad \text{on } \Gamma_{\theta}.
\end{aligned}
\tag{12}
$$

- Multiplying by w and integrating over Ω, the **general weak form** is:

General Weak Form — 1D Problems

Find $\theta(x) \in U$ such that

$$
\int_{\Omega} \frac{\mathrm{d}w}{\mathrm{d}x} A\kappa \frac{\mathrm{d}\theta}{\mathrm{d}x}\,\mathrm{d}x - \int_{\Omega} wf\,\mathrm{d}x - \left[wA\left(\bar{\Phi} - \beta(\theta - \bar{\theta})\right)\right]\Big|_{\Gamma_{\Phi}} = 0 \qquad \forall w \in U_0. \tag{13}
$$

Next: Approximation of Trial Solutions for 1D Problems

Thanks for your attention!